Amazon Elastic Load Balancing — Important Interview Questions & Answers

1.What is Elastic Load Balancing?
Elastic Load Balancing is an AWS service that distributes incoming traffic across multiple healthy targets such as EC2 instances, containers, and IP addresses. It helps improve application availability, scalability, and reliability.
Simple ga gurthupettuko:
ELB ante incoming traffic ni multiple healthy targets ki distribute chese AWS service.
Eng: Elastic Load Balancing distributes incoming traffic across multiple healthy targets.

2.What is a Load Balancer?
A Load Balancer receives incoming traffic from users and distributes it across multiple backend targets. It helps prevent a single server from handling all the traffic.
Simple ga gurthupettuko:
Load Balancer ante incoming traffic ni multiple backend servers madhya distribute chese service.
Eng: A Load Balancer distributes incoming traffic across multiple backend targets.

3.What is an Application Load Balancer (ALB)?
An Application Load Balancer is designed for HTTP and HTTPS traffic. It works at the application layer and can route traffic based on paths, hostnames, HTTP headers, and other application-level information.
Simple ga gurthupettuko:
ALB ante HTTP and HTTPS traffic ni application level lo route chese Load Balancer.
Eng: An Application Load Balancer routes HTTP and HTTPS traffic at the application layer.

4.What is a Network Load Balancer (NLB)?
A Network Load Balancer is designed for TCP, UDP, and TLS traffic. It works at the transport layer and is suitable for applications that require high performance and large numbers of connections.
Simple ga gurthupettuko:
NLB ante TCP, UDP and TLS traffic ni high performance tho handle chese Load Balancer.
Eng: A Network Load Balancer handles TCP, UDP, and TLS traffic with high performance.

5.What is a Gateway Load Balancer (GWLB)?
A Gateway Load Balancer is used to deploy, scale, and manage virtual network appliances such as firewalls and intrusion detection systems. It helps route network traffic through these security appliances.
Simple ga gurthupettuko:
GWLB ante firewalls and other network security appliances kosam use chese Load Balancer.
Eng: A Gateway Load Balancer is used to deploy and scale virtual network appliances.

6.What is the difference between ALB and NLB?
ALB works at Layer 7 and is mainly used for HTTP and HTTPS traffic with application-level routing. NLB works at Layer 4 and is mainly used for TCP, UDP, and TLS traffic with high performance.
Simple ga gurthupettuko:
ALB Layer 7 HTTP/HTTPS kosam, NLB Layer 4 TCP/UDP/TLS kosam use chestham.
Eng: ALB works at Layer 7, while NLB works at Layer 4.

7.What is the difference between ALB and GWLB?
ALB is mainly used for HTTP and HTTPS application traffic and provides application-level routing. GWLB is used to deploy and scale virtual network appliances such as firewalls and security appliances.
Simple ga gurthupettuko:
ALB application traffic kosam, GWLB security and network appliances kosam use chestham.
Eng: ALB handles application traffic, while GWLB handles virtual network appliances.

8.What is a Target Group?
A Target Group is a collection of targets that receive traffic from a Load Balancer. Targets can include EC2 instances, IP addresses, containers, and other supported resources.
Simple ga gurthupettuko:
Target Group ante Load Balancer traffic receive chese backend targets collection.
Eng: A Target Group is a collection of targets that receive Load Balancer traffic.

9.What is a Listener?
A Listener checks for incoming connection requests on a configured protocol and port. It then forwards the request to the appropriate target group based on the listener configuration.

Simple ga gurthupettuko:
Listener ante specific port and protocol meeda incoming traffic ni listen chese component.
Eng: A Listener receives incoming traffic on a configured port and protocol.

10.What are Listener Rules?
Listener Rules define how incoming requests should be routed to target groups. For an Application Load Balancer, rules can use conditions such as path, hostname, HTTP headers, or query parameters.
Simple ga gurthupettuko:
Listener Rules ante incoming request ni ye target group ki pampalo decide chese rules.
Eng: Listener Rules decide how incoming requests should be routed.

11.What are Health Checks?
Health Checks continuously check whether registered targets are healthy and ready to receive traffic. If a target becomes unhealthy, the Load Balancer stops sending new traffic to that target.
Simple ga gurthupettuko:
Health Check ante backend target healthy ga unda leda check chese process.
Eng: Health Checks determine whether targets are healthy and ready to receive traffic.

12.What happens if a target becomes unhealthy?
The Load Balancer health check identifies the target as unhealthy. It stops sending new traffic to that target and sends traffic to other healthy targets.
Simple ga gurthupettuko:
Oka target unhealthy ayite Load Balancer daniki traffic pampinchakunda healthy targets ki pampistundi.
Eng: If a target becomes unhealthy, the Load Balancer sends traffic to healthy targets.

13.What are Load Balancer Attributes?
Load Balancer Attributes are configuration settings that control the behavior of a Load Balancer. They can be used to configure features such as access logs, connection settings, and other load balancing options.
Simple ga gurthupettuko:
Load Balancer Attributes ante Load Balancer behavior ni control chese settings.
Eng: Load Balancer Attributes control the behavior and features of a Load Balancer.

14.What are Security Groups in Load Balancing?
Security Groups act as virtual firewalls that control inbound and outbound traffic for supported Load Balancers. We can allow required traffic based on ports, protocols, and source IP addresses.
Simple ga gurthupettuko:
Security Group ante Load Balancer ki incoming and outgoing traffic ni control chese firewall.
Eng: Security Groups control allowed inbound and outbound traffic for supported Load Balancers.

15.How does a Load Balancer work?
The client sends a request to the Load Balancer. The Load Balancer receives the request, checks the listener configuration, and forwards it to a healthy target in the target group.
Simple ga gurthupettuko:
Client → Load Balancer → Listener → Target Group → Healthy Target.
Eng: A Load Balancer receives client requests and forwards them to healthy targets.

16.Why do we use a Load Balancer?
We use a Load Balancer to distribute incoming traffic across multiple backend targets. It helps improve application availability, scalability, and reliability.
Simple ga gurthupettuko:
Multiple servers madhya traffic distribute chesi application availability and scalability improve cheyyadaniki Load Balancer use chestham.
Eng: We use a Load Balancer to improve application availability, scalability, and reliability.

17.Can a Load Balancer work across multiple Availability Zones?
Yes, a Load Balancer can distribute traffic across targets located in multiple Availability Zones. This helps improve application availability and fault tolerance.
Simple ga gurthupettuko:
Load Balancer multiple AZs lo unna targets ki traffic distribute cheyyachu.
Eng: A Load Balancer can distribute traffic across multiple Availability Zones.

18.What is Cross-Zone Load Balancing?
Cross-Zone Load Balancing allows a Load Balancer to distribute traffic across targets in different Availability Zones. It helps distribute traffic more evenly across available targets.
Simple ga gurthupettuko:
Cross-Zone ante different AZs lo unna targets madhya traffic balance cheyyadam.
Eng: Cross-Zone Load Balancing distributes traffic across targets in different Availability Zones.

19.What happens when all targets in a Target Group are unhealthy?
If all targets are unhealthy, the Load Balancer does not have healthy targets available to receive normal traffic. This can cause the application to become unavailable depending on the configuration.
Simple ga gurthupettuko:
All targets unhealthy ayite healthy server lekapovadam valla application unavailable avvachu.
Eng: If all targets are unhealthy, the application may become unavailable.

20.What is SSL/TLS termination?
SSL/TLS termination means the Load Balancer handles the client HTTPS connection and decrypts the traffic before forwarding it to the backend according to the configured setup. This reduces the work required from backend servers for handling client-side TLS connections.
Simple ga gurthupettuko:
SSL/TLS termination ante client HTTPS connection ni Load Balancer handle chesi backend ki traffic forward cheyyadam.
Eng: SSL/TLS termination allows the Load Balancer to handle the client HTTPS connection.

21.What is the difference between Load Balancer and Target Group?
A Load Balancer receives incoming traffic and distributes it to registered targets. A Target Group is the collection of backend targets that receive the traffic from the Load Balancer.
Simple ga gurthupettuko:
Load Balancer traffic receive and distribute chestundi, Target Group backend targets collection.
Eng: A Load Balancer distributes traffic, while a Target Group contains the backend targets.

22.What is the difference between Listener and Listener Rule?
A Listener receives incoming traffic on a configured port and protocol. Listener Rules decide where that traffic should be forwarded based on configured conditions.
Simple ga gurthupettuko:
Listener traffic ni receive chestundi, Listener Rule traffic ekkadiki vellalo decide chestundi.
Eng: A Listener receives traffic, while Listener Rules decide how to route it.

23.How would you troubleshoot a Load Balancer issue?
I would check the Load Balancer status, listener configuration, target group, health checks, Security Groups, NACLs, routing, and application logs. I would also verify whether the registered targets are healthy.
Simple ga gurthupettuko:
Load Balancer problem vaste Listener → Target Group → Health Check → Security Group → NACL → Logs check chestham.
Eng: For a Load Balancer issue, I would check listeners, target groups, health checks, security rules, routing, and logs.

24.How do you create an Application Load Balancer?
First, we create the Application Load Balancer and select the required VPC and subnets. Then we create a target group, register targets, configure a listener, and add listener rules if required.
Simple ga gurthupettuko:
ALB create chesi VPC/Subnets select chesi Target Group, Targets and Listener configure chestham.
Eng: We create an ALB, target group, targets, and listener to route application traffic.

25.What is the most important thing to remember about Elastic Load Balancing?
Elastic Load Balancing distributes incoming traffic across healthy backend targets to improve application availability, scalability, and reliability. The important concepts are ALB, NLB, GWLB, Target Groups, Listeners, Listener Rules, Health Checks, Attributes, and Security Groups.
Simple ga gurthupettuko:
ELB ante incoming traffic ni healthy targets ki distribute chesi application availability and scalability improve chese service.
Eng: Elastic Load Balancing distributes traffic across healthy targets to improve application availability and scalability.

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